

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY 2021 TRIMESTER

FINAL ASSESSMENT

UCCN1004 DATA COMMUNICATIONS AND NETWORKING

MONDAY. 13 SEPTEMBER 2021

TIME : 8.30 AM – 11.30 AM (3 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING

Instructions to Students:

REMINDER: You are reminded to read and adhere to the Final Assessment Instructions to Candidates that has been made available through the UTAR Portal before the commencement of Final Assessment (FA). The detailed instructions for this FA are as follows:

General

1. This Final Assessment (FA) is an Individual, time restricted assessment which consists of FOUR (4) questions. Each question carries 25 marks.
2. You are required to answer **ALL** questions, and submit the **ANSWER SCRIPT** by **11.30 am, 13 SEPTEMBER 2021**.
3. During the period of 3 hours of this FA, the examiner(s) can be reached at
 - (a). **Zoom:** [Click here to join chat](#)
 - (b). **WhatsApp** at 017-5118632
 - (c). **MS Team:** aunyc@utar.edu.my

You may use the above e-platform(s) to check with the examiner(s) if you need any clarification on this FA question paper

4. You may refer to any books, lecture notes, published materials, online resources, etc when answering the questions. Candidates are reminded that **Copy-and-Paste, Consultation, Discussion and Sharing of Answers** with others are **STRICTLY PROHIBITED** in this FA.

UCCN1004 DATA COMMUNICATIONS AND NETWORKING**Answer Script File**

5. The answer script **MUST** be submitted as a PDF file, in A4 size format.

Note: File size restrictions of maximum 100MB applied.

6. Please check you index number generated by the Division of Examinations, Awards, and Scholarships (DEAS). You **MUST** name your answer script using the following file name for submission:

UCCN1004_FA_[Programme Abbreviation]_[Your Index Number]

For example, if you are from the degree programme CN, and your Index Number is A99999CBCNF, then your answer script should be named as:

UCCN1004_FA_CN_A99999CBCNF.pdf

Answer Script Submission

7. Your answer script file has to be submitted to the following platform(s) before the due time/date.

- (a). Attach your answer script at Google Form. [Click here to upload your answer script.](#)

(Use your "1UTAR" email account to access the Google Form. Only one submission is allowed)

- (b). Send your answer script to the following Email Account: aunyc@utar.edu.my using your 1UTAR email account. Please use the file name of your answer script. For example,

UCCN1004_FA_[Programme Abbreviation]_[Your Index Number]

8. Please make sure you submit the correct, complete and final version of your answer script and the same copy of answer script file is to be submitted to the above platform(s). If different answer scripts are received, the examiner will randomly choose either one of them to mark.

UCCN1004 DATA COMMUNICATIONS AND NETWORKING**Contents of Answer Script**

9. The **first page** of your submission is the **Final Assessment Cover Page**. You **MUST** use the template given and fill in the following information
 - Your Degree Programme (Abbreviation)
 - Your Index Number
 - Your Name
 - Your Student ID
10. The **second page** of your submission is the **Final Assessment Declaration Statement**. You **MUST** use the template given, and digitally sign on the form to indicate the authenticity of your submitted work is without plagiarism.
11. Each question should be answered starting on a new page. There are no restrictions of page number and words count.
12. For answer scripts that have text-based answers only, all texts **MUST** be typed and recommended using Times New Roman characters with font size no less than 12, except for the drawings and equations/calculations. It is recommended that you embed drawings into your answer script.
13. For the drawings, you can draw/write them on a blank paper, and then take snapshots of the drawings and insert them into the answer script (Word document). Alternatively, you can edit directly on the templates given in the link in some of the questions and insert the final drawing to the answer script.
14. Please include a page number on **each and every page** of your answer script. Ensure that each page of answer scripts is in sequence prior to online submission.

WARNING OF PLAGIARISM

15. Answer scripts shall be uploaded to Turnitin to check for the originality of submitted answers. In the case of suspected plagiarism, the evidence will be submitted to the Examination Disciplinary Committee of the University. Disciplinary action shall be taken against any candidate who is found to have plagiarized in the answer submitted. Hence, candidates are reminded to abide by all University Rules and Regulations and any instructions/guidelines relating to examinations/assessments.

UCCN1004 DATA COMMUNICATIONS AND NETWORKING**Questions: [Total: 100 Marks]**

Q1. Answer the following questions based on the network shown in Figure 1.

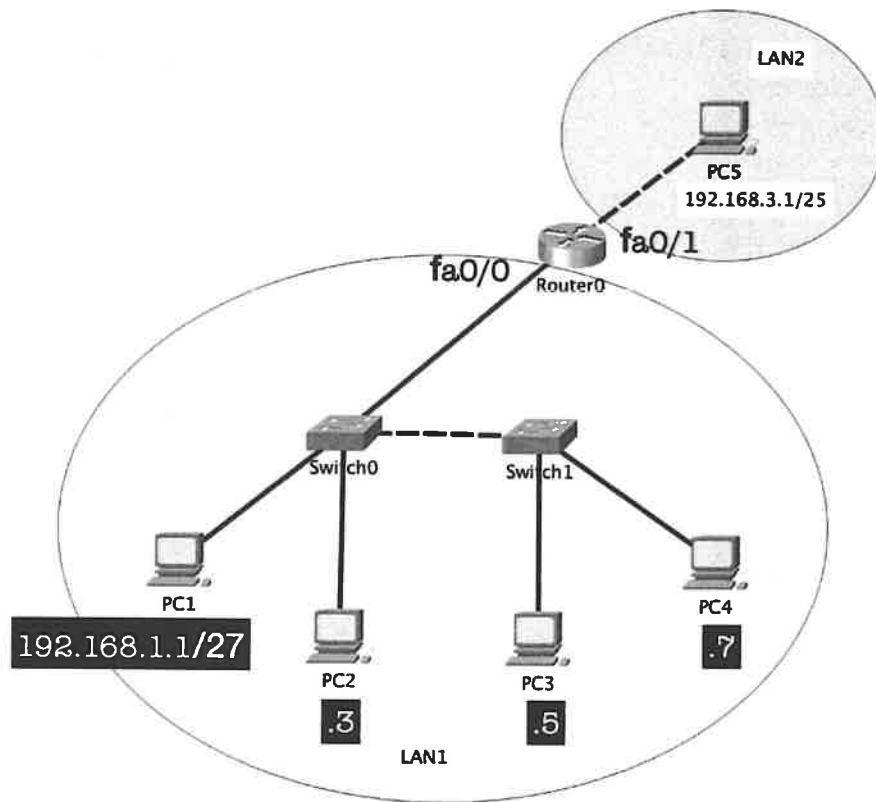


Figure 1

- (a) Based on the LAN1 in Figure 1, calculate the following:
- The Network Address (2 marks)
 - The usable IP address range (2 marks)
 - The maximum number of usable Host IP Address (2 marks)
 - The remaining number of usable Host IP Address (2 marks)
- (b) Design and place an Access Control List (ACL) on Router0 to prevent PC1 and PC2 from communicating with PC5 while PC3 and PC4 can still communicate with PC5. The additional requirements are:
- use a blacklist approach
 - use a wildcard mask to define the source IP range
 - use '1' as the ACL number
 - place access-list 1 in fa0/0 on Router0
- (12 marks)

UCCN1004 DATA COMMUNICATIONS AND NETWORKING**Q1. (Continued)**

- (c) State the path of ICMP_request packet when PC2 tries to ping PC3. (2 marks)
- (d) The network admin found that routing is not configured on Router0. However, he later discovered that PC2 managed to ping PC5. Explain why? (3 marks)
- [Total : 25 marks]

Q2. Figure 2 shows some devices that are connected to Switch S0 in a local area network.

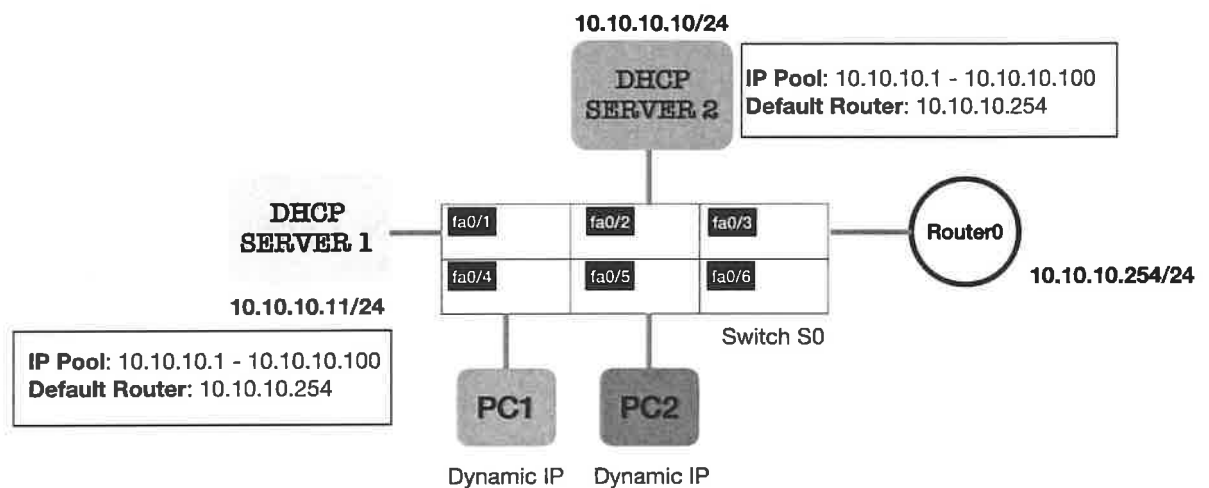


Figure2

- (a) Supposed that PC1 obtains a dynamic IP from 'DHCP Server 1'. The IP address '10.10.10.1' has been offered to PC1. Illustrate the FOUR types of DHCP messages (DORA) between PC1 and the 'DHCP Server 1' during this process. (8 marks)
- (b) Based on the DHCP addressing architecture in Figure 2, explain HOW it is possible where PC1 and PC2 are both assigned the same IP address. (5 marks)

UCCN1004 DATA COMMUNICATIONS AND NETWORKING**Q2. (Continued)**

- (c) The network admin has decided to disable both DHCP servers and configure Router0 to assign IP addresses instead. **State** the Cisco IOS command to set Router0 as the DHCP server for this network using the following parameters (*ignore router prompt*). (8 marks)

Pool Name: avoSauce
IP Pool: 10.10.10.1 - 10.10.10.253
Default Router: 10.10.10.254
DNS Server: 10.10.10.18

- (d) The IP range **10.10.10.10 – 10.10.10.20** are reserved for servers use. State the Cisco IOS commands on Router0 to exclude the IP address range from the IP address pool. (4 marks)

[Total : 25 marks]

- Q3. Figure3 shows the network topology in the Apple Campus. DuaLipa is a secret agent sent to hack the **Apple Core Server** in the network.

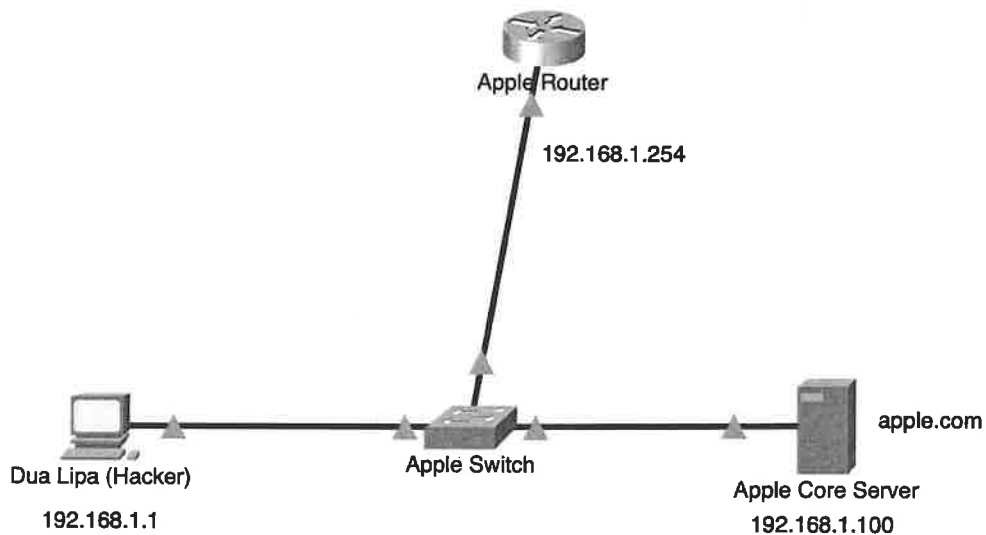


Figure3

- (a) DuaLipa uses NMAP to perform a SYN scan on Apple Core Server and found that the FTP service is opened. Visualize the flow of SYN scan for this transaction using a diagram.

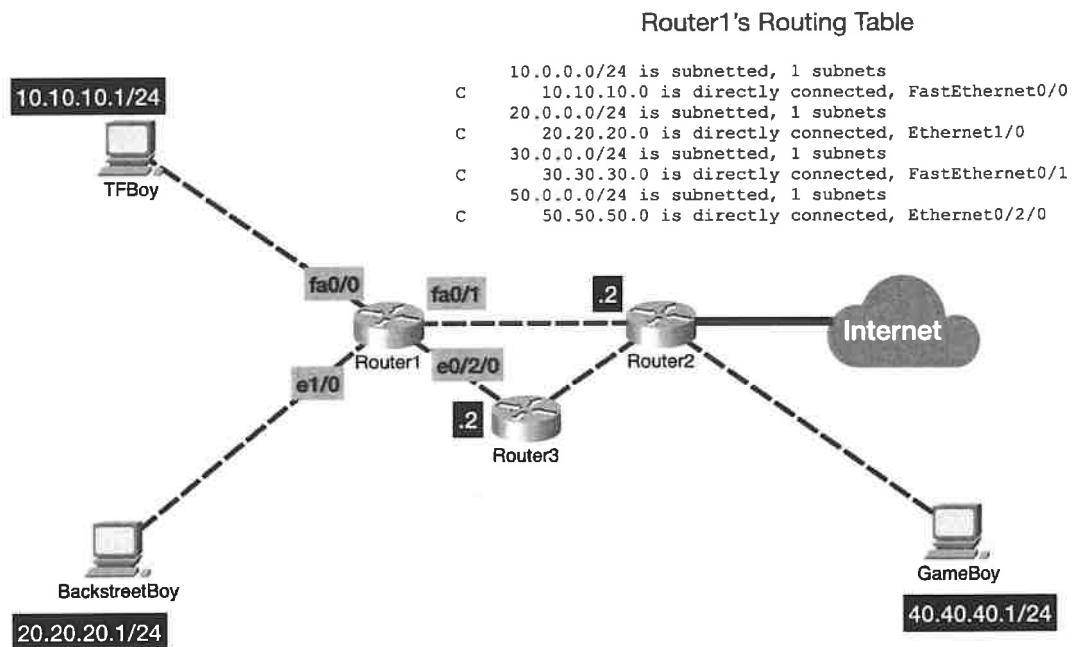
(10 marks)

UCCN1004 DATA COMMUNICATIONS AND NETWORKING**Q3. (Continued)**

- (b) Explain why the SYN scan in NMAP is designed to be half-opened in the context of scanning speed? (6 marks)
- (c) DuaLipa performed a ping scan using the command '`nmap -sp 192.168.1.1/24`'. The scan results returned only one IP 192.168.1.254. Suggest **ONE (1)** possible reason why the *Apple Core Server* IP is not listed? (4 marks)
- (d) The firewall has been configured using a white-list approach to prevent NMAP probes. The network admin later found that the ping scan is still possible in the network during the penetration testing. Suggest **ONE (1)** possible reason to explain why ping scan still works? (5 marks)

[Total : 25 marks]

Q4. Figure 4 shows a fully converged network topology.

**Figure 4**

- (a) Suppose that Router1, Router2, and Router3 are all configured with RIPv1. State the path the ICMP packet would take when *TFBoy* pings *GameBoy*. Explain your answers. (5 marks)
- (b) *TFBoy* received an ICMP error message when trying to ping the *Gameboy*. Identify which device generated this error message? (5 marks)

UCCN1004 DATA COMMUNICATIONS AND NETWORKING**Q4. (Continued)**

- (c) State the Cisco IOS command to configure a default route to the Internet on Router1. (5 marks)
 - (d) Configure routing table in Router1 to 'force' all traffic from TFBoy to GameBoy to always take Router1->Router3->Router2. (5 marks)
 - (e) Discuss how do RIPv1 routers forward packets if there are two path with the same hop count to the same destination network (5 marks)
- [Total : 25 marks]
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